D3: Alternatives, Selection Rationale and Storyboard

Ideate Stage (Weeks 9–10) • Team G10 – MSM Technologies

Item	Detail
Team	G10, MSM Technologies
Section	G
Stage	Ideate (Weeks 9 to 10)
Deliverable	D3, due Week 9
Project	Usability of the FAST-NU Insignia library portal: study room booking
Submitted by	Mirza Yahya Adeel (23I-0768), coordination lead for Ideate
Input from D2	Requirements R1 to R5, POV1 to POV3, persona spectrum A to C, scenario and journey map. Evidence IDs (e.g. P09-B1, SV-5) are the D2 IDs.

Team members

Name	Roll Number	Role (from the D0 charter)
Muhammad Umar	23I-0782	Team member; coordination lead for Prototype
Mirza Yahya Adeel	23I-0768	Team member; coordination lead for Empathize and Ideate
Muhammad Usman Al Haq	23I-0585	Team member; coordination lead for Define
Shahmeer Atif	23I-0711	Team member; coordination lead for Test

D2 requirements we designed against

Our scope is the core task from D2: booking a study room through FAST-NU Insignia. The requirements and success criteria below come from D2; the criteria are shortened here. Alternative IDs carry the author's initials: MY = Mirza Yahya Adeel, MU = Muhammad Umar, UH = Muhammad Usman Al Haq, SA = Shahmeer Atif.

ID	Type	Requirement (outcome)	Success criterion (short form)
R1	Usability	First-time bookers reach the room-booking page unassisted.	At least 80% reach it within 2 minutes from the portal address, on their own phone (n of at least 10).
R2	Usability	Students can tell which rooms are free for a chosen day and time from the booking page alone.	At least 90% identify every free and every taken room and slot in three prepared views, at most 30 seconds per view, no help.
R3	UX	Students feel sure a booking they made is held for them.	Mean of at least 4.0 on "I am sure this booking is held for me" (1 to 5), no rating below 3.
R4	Usability	Students enter the intended date, slot and length correctly on the first attempt.	At least 90% book a specified room, date and two-hour slot with no correction and no help.
R5	Accessibility	Status can be read and booking completed on a small phone without colour or zooming.	All 6 checkpoints pass at 320 CSS px and 200% text: no two-way scrolling, no pinch-zoom, targets at least 24 by 24 px, status without colour, contrast 4.5 to 1, screen reader announces status and result.

On 22 October 2026 we also took screenshots of the live portal, on a desktop browser and on a phone browser, to check that the problems we were designing against are still there. They are referred to in Part 4 and collected in Appendix A. Anything in the storyboard that is not in those screenshots is our proposal, not an existing Insignia feature.

Part 1 | Individually generated alternatives

1.1 How we ran the charrette

We did the charrette on campus on Monday 19 October 2026, in a free slot after our lab. Each of us worked alone for 20 minutes with phones away, and no talking until all four were done. The only things on the table were our D2 outputs: persona spectrum, journey map, POV statements, HMW questions and the requirement list.

Five concepts each, one per sheet, and you had to write which requirement(s) it hits before looking at anyone else's sheets. The one rule was that your five had to be different approaches (what changes, who does it, or which channel), not the same screen five times with small tweaks. Some of us wrote full sentences and some just wrote notes, so the concepts below are typed up close to how they were written on the sheets, with only spelling and obvious shorthand fixed.

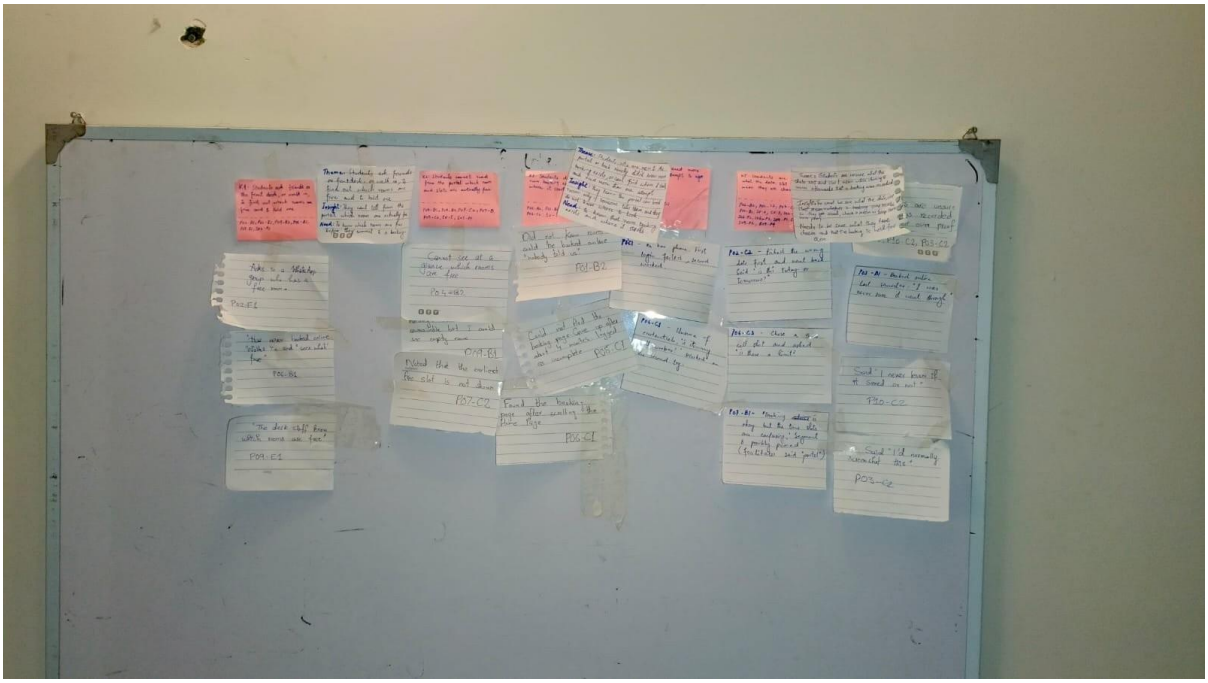


Figure 1. Our D2 affinity board, which we ideated from. Pink cards are the theme, insight and need statements; the white notes under them are the evidence notes, labelled with the D2 evidence IDs (e.g. P02-E1, P04-B2, P09-B1, P07-C2) that the concepts and critique below refer to.

1.2 Mirza Yahya Adeel (MY)

ID	Concept	D2 requirement(s) addressed
MY-1	**Availability-first booking.** Booking page opens straight on today's rooms. Under each room, every free start time is its own button ("Free 11:00", "Free 12:00"...). You start from a time that's already free, not from a date box and the slider.	R2 (see free rooms on the page itself), R4 (no slider, no separate date step)
MY-2	**Floor-plan map.** Simple drawing of the study rooms, each one labelled Free/Taken plus a pattern so it isn't only colour. Tap a room to get its free times.	R2, R5
MY-3	**QR code on each room door.** Scan it, see that room's schedule for today, book it right there.	R2 if you're already in the library (P09-B1). Bit of R1?
MY-4	**Booking pass.** After booking you get a "pass" screen: room, date, start–end time, booking ID. Stays under "My bookings", and there's a share button for the group chat. Basically so nobody has to screenshot the confirmation anymore (P03-C2, P10-B2).	R3
MY-5	**WhatsApp bot.** Message a library number, it replies with the free rooms, reply with a number to book.	R1 (nothing to find in the portal), R2

1.3 Muhammad Umar (MU)

ID	Concept	D2 requirement(s) addressed
MU-1	**"Book a study room" on home page.** Big tile, first thing after login. Also a link on the login page itself, takes you there after sign-in.	R1
MU-2	**Next free room button.** Pick length, it just offers one: "Room 3, 11:00–13:00. Book it?" Yes / show next.	R4, some R2 (P07-C2)
MU-3	**Screen at library entrance.** Shows rooms free now + next. Same thing on a public web page, no login.	R2
MU-4	**Step-by-step booking.** One question per screen. Day (Today / Tomorrow, with the	R4, R5 (big buttons)
ID	**Concept**	**D2 requirement(s) addressed**
	date written out), then start time, then length as buttons: 1 hr, 2 hr, 3 hr. No slider.	
MU-5	**Room watch.** Nothing free: "Tell me when a room frees up", then a notification.	R2 partly

1.4 Muhammad Usman Al Haq (UH)

ID	Concept	D2 requirement(s) addressed

UH-1	**Check-in and auto-release.** I think a lot of the "room shows booked but it's empty" problem (P09-B1, SV-5) is people booking and not turning up. So: you have to check in within 15 minutes of your start time, either at the desk or by scanning a code on the door. If you don't, the booking is cancelled and the room shows as free again. Not really a screen change, more of a rule change.	R2 (whether the status is actually correct, not how it's shown). Also the double booking in S05-F1.
UH-2	**Fill-in-the-sentence booking.** Instead of a form, the student completes "I need a room for ___ people on ___ from ___ for ___ hours" and the portal answers with the rooms that fit. This also finally asks for group size, which P10 pointed out the form never does (P10-B1).	R4, partly R2
UH-3	**Email + SMS receipt.** Every booking sends an email and an SMS with room, date, time, a calendar file (.ics) and a cancel link, so there's proof outside the portal too.	R3
UH-4	**Desk uses the same system.** When students book at the desk, the staff should book through the same portal schedule for them, so the desk and the portal never disagree and the student still ends up with a record. Persona A is going to keep going to the desk anyway, so this helps them.	R2, R3
UH-5	**Weekly repeat booking.** For people who book the same room at the same time every week (persona C, P04-A1). Set it up once and it renews until you cancel it.	R2 + R4 partly, persona C only

1.5 Shahmeer Atif (SA)

ID	Concept	D2 requirement(s) addressed
SA-1	**Timeline grid.** Rooms = rows, hours = columns. Free cells outlined + "Free", taken ones hatched. Drag across free cells to pick your time.	R2, R4 (kind of)
SA-2	**First-login guide.** 3-step overlay pointing at room booking the first time you sign in. Orientation-week email links straight to it. ("nobody told us", P01-B2)	R1
SA-3	**Kiosk.** Tablet at the library entrance, log in with roll no., book from today's free rooms.	R1 + R2 (only if you're there)
SA-4	**Group link.** Organiser books, sends link, everyone in the group sees room + time, and it shows in their own bookings list too.	R3
SA-5	**Home-screen widget.** Shows rooms free in the next 2 hrs without opening the portal / logging in again.	R2, maybe R1

Part 2 | Team convergence

2.1 How the critique ran

We met on campus on Tuesday 20 October 2026 for about 90 minutes, all four of us, with the Ideate lead facilitating. All 20 sheets went up on one wall and we worked through them as follows.

- **Silent read.** Everyone read all 20 sheets before anyone said anything.
- **Critique of every alternative.** For each sheet in turn we agreed one strength, one concern and the requirement it best satisfies, each tied to D2 evidence or a D2 success criterion. The author spoke last, so the discussion didn't turn into people defending their own ideas.
- **Coverage scoring.** We rated each alternative against the R1 to R5 success criteria and noted whether we could realistically prototype and test it in D4 and D5.
- **Shortlist and comparison.** Anything with at least one Full rating and a testable form went forward. The ones aimed at our leading requirement, R2, were compared head to head.
- **Decision.** By majority vote. Because the front-runner (MY-1) was the facilitator's own concept, Shahmeer (Test lead) chaired the vote instead, and Mirza voted last.

Grouping by approach. Before the critique we sorted the 20 sheets by what they change. We did this mainly to check that we had actually diverged, rather than written 20 versions of the same screen.

Approach family	Alternatives	What it changes
F1. Change the booking page itself	MY-1, MY-2, MU-2, MU-4, UH-2, SA-1	How rooms, times and length are shown and chosen
F2. Make booking easy to find	MU-1, SA-2, SA-5	How students reach booking
F3. Prove the booking is held	MY-4, UH-3, SA-4	What students get after booking
F4. Bring availability into the building	MY-3, MU-3, SA-3	Physical displays and devices
F5. Change the process or channel	MY-5, UH-1, UH-4, UH-5, MU-5	Policy, staff workflow, messaging, notifications

One thing became clear at this point: no single family covers R1, R2, R3 and R4 together. F1 is the only family that reaches R2 and R4, F2 the only one aimed at R1, and F3 the only one aimed at R3.

2.2 Critique of all 20 alternatives

ID	Strength	Concern	Best satisfies
MY-1	Free times are the first thing seen, at the exact step where students give up on the portal and ask people instead (journey stage 3; P09-B2, P04-E1).	With many rooms the list gets long. Whether a time shown as free is really free still depends on the portal's data (P09-B1).	R2
MY-2	Matches how students already check rooms, by looking at them (P06-B1, "sees what's free").	Needs room positions we haven't mapped. At 320 px, small rooms become tiny targets (R5 checkpoint 3).	R2 (larger screens)
MY-3	Works where students already are, and the status can be checked against the real room.	Only helps students already inside the library. Morning bookers working from elsewhere (P04-B1; P09-B1 at 9 am) get nothing. Needs printed codes on doors.	R2 partly
MY-4	Replaces the screenshots, sticky notes and desk checks students already rely on (P03-C2, P10-B2, P03-B2) with one record.	A pass only reassures if the booking really was saved. It does nothing before the booking is made.	R3
MY-5	Uses the channel students already use to find rooms (P02-E1, P10-E1).	Moves booking out of the portal. Text size and screen reader output are outside our control (R5 checkpoint 6). Hard to test against R2's three prepared views.	R1 partly
MU-1	Directly removes the hunt for the page that took P01 about three minutes and made P05 give up after about four (P01-C2, P05-C1).	Only fixes the first step. Once on the booking page, the old problems are still there.	R1
MU-2	Fastest path for practiced users, and answers P07-C2 (earliest free slot not shown).	Hides which other rooms are free, so a student can't identify every free and every taken room, which R2's criterion requires. A group that wants a particular room loses the choice.	R4
MU-3	Visible to everyone without sign-in, including persona A students who never open the portal.	Hardware and an always-on public page are beyond what we can build. It shows rooms but books nothing.	R2 partly
MU-4	Removes the slider that three survey respondents complained about (S02-F2, S06-F1, S09-F1) and writes the date out in words (P02-C2, "is this today	More screens means more steps. On its own it still doesn't show which rooms are free before the student commits.	R4
ID	**Strength**	**Concern**	**Best satisfies**
	or tomorrow?").		
MU-5	Removes the refreshing that practiced users do (P04-B1).	Only useful once the student has already found no free room. Needs a notification channel the portal may not have.	R2 partly
UH-1	The only alternative that goes after why rooms show the wrong status (P09-B1, SV-5, S05-F1).	A library policy and back-end change, not an interface. We can't build or test it in D4 or D5, and it needs the library administrators to agree.	R2 (accuracy, not readability)
UH-2	Captures group size, which P10 noticed the form never asked for (P10-B1).	Several typed or picked fields on a phone is exactly where mis-taps happened (P08-C2). We don't know whether a capacity rule even exists.	R4 partly
UH-3	Gives students a record outside the portal that they can find later.	Depends on email and SMS integration, and arrives after the moment of doubt at the confirmation screen (P03-C2).	R3 partly
UH-4	Accepts that persona A will keep using the desk (P09-E1, P10-A1) and keeps desk and portal on one schedule.	Mostly a staff-side change. It doesn't change what students see on the booking page, which is what R2 tests.	R2 partly
UH-5	Fits the weekly pattern of persona C (P04-A1, P07-A1).	Helps the practiced bookers who already succeed and could tie up rooms for everyone else.	R2 partly (persona C)
SA-1	Shows every room and hour in one view; the best of the set for reading a whole day at a glance (P04-B2).	A rooms-by-hours grid at 320 px needs two-way scrolling, which fails R5 checkpoint 1. Drag selection brings back the click-and-undo trouble reported for the slider (S09-F2, S09-F3).	R2 (wide screens)

SA-2	Answers "nobody told us" (P01-B2) at the moment of first sign-in.	Overlays often get skipped, and it helps once, not the occasional user who has forgotten (P05-B1).	R1 partly
SA-3	A shared device for students without a phone or data, or who would rather book in person.	Hardware cost, a queue in exam weeks (P03-A1), and no help to students booking before they arrive.	R2 partly (on site)
SA-4	Replaces passing booking details around the group chat (P10-E1) with the record itself.	Adds group membership, which no participant asked for. The real doubt is the organiser's own (P03-B1).	R3 partly
SA-5	Free rooms visible without signing in again, avoiding the sign-in retries we saw in P01-C1 and P06-C1.	Needs an app or installable portal. Insignia is a thirdparty system, so this may not be possible.	R2 partly

2.3 Requirement coverage of all 20 alternatives

Full: we think the concept on its own could plausibly meet that requirement's D2 success criterion. **Part:** it helps, but wouldn't meet the criterion alone. **Blank:** not addressed. These are our predictions going into D4, not test results.

ID	R1	R2	R3	R4	R5	Can we prototype and test it in D4/D5?
MY-1		Full		Full	Part	Yes
MY-2		Full			Part	Yes
MY-3	Part	Part				No (printed codes)
MY-4			Full			Yes
MY-5	Part	Part				Partly
MU-1	Full					Yes
MU-2		Part		Full	Part	Yes
MU-3		Part				No (hardware)
MU-4				Full	Part	Yes
MU-5		Part				Yes
UH-1		Part				No (library policy)
UH-2		Part		Part		Yes
UH-3			Part			Partly
UH-4		Part	Part			No (staff system)
UH-5		Part		Part		Yes
SA-1		Full		Part	Fails cp. 1	Yes
SA-2	Part					Yes
ID	**R1**	**R2**	**R3**	**R4**	**R5**	**Can we prototype and test it in D4/D5?**
SA-3	Part	Part				No (hardware)
SA-4			Part			Yes
SA-5	Part	Part				Partly

Reading the matrix. MY-1 is the only alternative with two Full ratings, and one of them is R2, the requirement tied to our leading POV1. MU-1 is the only Full on R1 and MY-4 the only Full on R3. Nothing is Full on R5, because we haven't yet checked any concept against the six R5 checkpoints; we carry R5 as an open risk. MY-1 is rated Part on R5 rather than Full: its single column removes two-way scrolling (checkpoint 1), but the other five checkpoints are still unchecked.

2.4 Shortlist and head-to-head comparison

Our shortlist rule was simple: at least one Full rating, and something we can actually prototype and test. Seven alternatives made it through: MY-1, MY-2, MY-4, MU-1, MU-2, MU-4 and SA-1.

Choosing the core. The core had to address R2, because R2 carries the leading opportunity. Four shortlisted alternatives touch R2, so we compared them on the R2, R4 and R5 criteria.

Alternative	R2: identify every free and taken room in three views, 30 s each	R4: correct date, slot and length first time	R5 at 320 px	Result
MY-1 Availability-first	Yes. Every room is listed with its free times in words.	Yes. Day and start chosen in one tap; no slider.	Single column; time buttons wrap. No two-way scroll.	**Selected as core**
SA-1 Timeline grid	Yes, on a wide screen.	Partly. Drag to select repeats the slider errors (S09-F2, S09-F3).	Fails checkpoint 1: grid needs two-way scrolling.	Runner-up, not selected
MY-2 Floor-plan map	Yes, if room labels fit.	No. Needs a separate time step.	Small room targets; layout not yet mapped.	Not selected
MU-2 Next free room	Partly. It shows the earliest free room, but only one room at a time, so it can't meet the "every free and every taken room" criterion on its own.	Yes.	Likely passes.	Not selected

Elements adopted from other alternatives. MY-1 covers R2 and R4 but, like every F1 alternative, does nothing for R1 or R3. So we added the single Full alternative from each of those columns. Both are small additions and neither changes MY-1's flow.

From	Element adopted	Why
MU-1	"Book a study room" tile first on the home page, and a link on the sign-in page	Only Full rating on R1
MY-4	Booking pass with booking ID, saved under My bookings, with Share	Only Full rating on R3
MU-4	Length chosen from fixed 1, 2 and 3 hour buttons (not MU-4's one-question-per-screen wizard)	Removes the slider from the length step (R4). We left out the full wizard because it adds screens and brings the date question back.

Decision. MY-1 as the core, with the three elements above. The vote was unanimous, 4 of 4. We discussed SA-1 as an alternative core and dropped it once we agreed on the 320 px check in the comparison table.

2.5 Who did what in D3

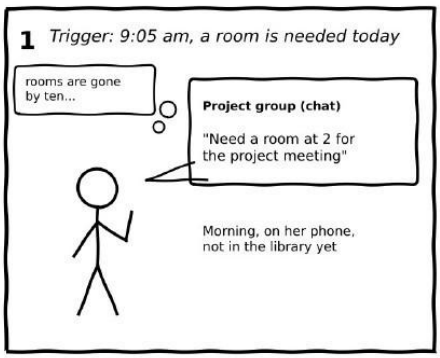
Member	Contribution
Mirza Yahya Adeel	Ideate lead: set up and facilitated the charrette and critique, five concepts (MY-1 to MY-5), compiled the report and Part 4 rationale
Muhammad Umar	Five concepts (MU-1 to MU-5), kept the critique record during the session, took the portal screenshots in Appendix A
Muhammad Usman Al Haq	Five concepts (UH-1 to UH-5), coverage scoring against the D2 success criteria
Shahmeer Atif	Five concepts (SA-1 to SA-5), chaired the selection vote, drew the storyboard panels

Part 3 | Storyboard

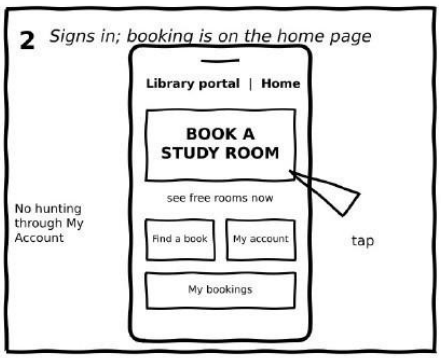
Persona: B, the hesitant portal booker (a composite of P02 and P03). She has used the portal before but is never sure it worked, so she usually backs it up with friends, the desk or a screenshot.

Scenario: the D2 core scenario. Her group needs a room for a meeting later the same day. It is morning, when rooms "disappear by ten in the morning" (P04-B1), and she is on her phone (P01-C1, P08-C1).

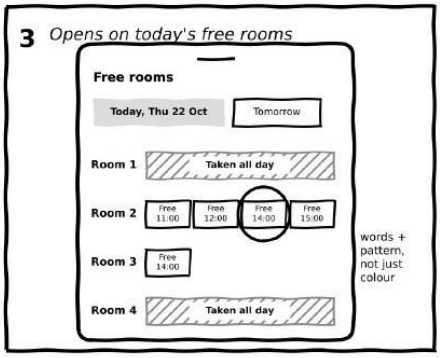
We kept the panels deliberately rough: they are about structure and flow, not visual design. Room names, times and the booking ID are placeholders, and the controls shown are what we plan to prototype in D4, not things Insignia does today.



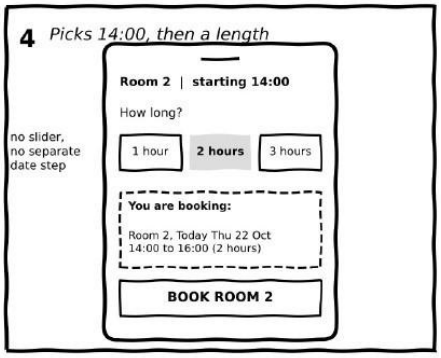
Trigger. 9:05 am, the project group needs a room at 2 pm. She expects rooms to be gone by ten.



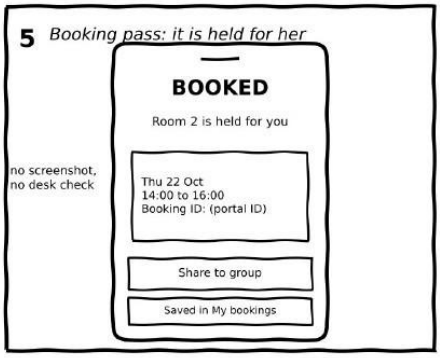
Reach booking. After sign-in, "Book a study room" is the first thing on the home page.



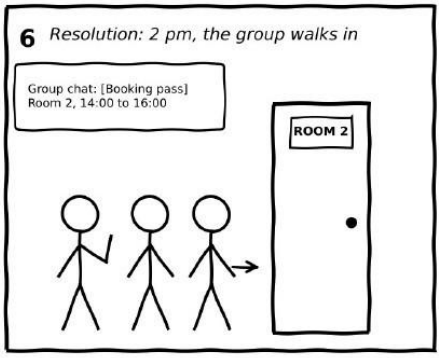
Key moment: see what is free. The page opens on today's rooms. Free times are buttons that say Free; taken rooms say Taken.



Key moment: choose. She taps 14:00, then "2 hours". One sentence states exactly what will be booked.



Confirm. A booking pass says Room 2 is held for her, with an ID. It is saved under My bookings.



Resolution. She shares the pass to the group. At 2 pm they walk into Room 2: no desk visit, no friend sent ahead.

If no room is free (not drawn). At panel 3, every room reads "Taken all day" and a line under the list says so in words, with a "Tomorrow" button and the earliest free time tomorrow. This is the state flagged in OR6 and will be designed in D4A.

Storyboard, six panels: (1) the group needs a room; (2) booking is the first thing on the home page; (3) today's rooms with Free/Taken labels; (4) she picks a start time and a length, and one sentence states what will be booked; (5) the booking pass; (6) she shares the pass and the group walks into the room.

Part 4 | Design rationale

4.1 Concept summary: Free Rooms First

Room booking is one tap from the portal home page and opens straight onto the day's rooms. Every free start time is a labelled button, so a student books by picking a time that is already shown as free, then a length from three fixed buttons. Before booking, one sentence states exactly what will be booked. Afterwards, a pass says the room is held and gives a booking ID; the pass stays under "My bookings" and can be shared with the group.

A note on naming: the current portal already has a menu entry called "My Room Bookings" (Appendix A, Figure A1).
In our design we shorten it to "My bookings", and that is the label used in the storyboard and in the rest of this report. We haven't been able to see what that page currently contains, so the pass and booking ID are new.

4.2 Requirements addressed, and how

Req.	How the concept addresses it	Panel	Status after D3

R1	"Book a study room" is the first tile on the home page after sign-in, and the sign-in page links to it (from MU-1). First-time bookers no longer need to know it sits under My Account → Booking → Book Room (Figure A1).	2	Addressed in design. Still to be tested with first-time bookers on their own phones.
R2	The booking page opens on today's rooms (Today and Tomorrow tabs). Each room lists its free start times as buttons reading "Free" and the time, earliest first (P07-C2); fully taken rooms read "Taken all day". Status never relies on colour alone (P09-C2). This replaces the current colour-coded month grid and narrow time strip, which have no visible text legend (Figure A2).	3	Addressed for reading the page. Not addressed for whether the portal's data is correct (risk OR1).
R3	Before booking, one sentence states room, date, start, end and length. After booking, a pass says "Room 2 is held for you" with a booking ID, is saved under My bookings and can be shared (from MY-4).	4, 5	Partly addressed. Depends on what the portal actually records on submit, which we never observed (OR3).
R4	Date and start are chosen together by tapping a free time, so there is no separate date picker and no time-range slider (both visible in Figures A2 and A3). Length is one of three buttons (from MU-4). The summary sentence lets the student catch a wrong entry before booking.	3, 4	Addressed in design. Length options depend on the library's real rules (OR2).
R5	One-column list; time buttons wrap onto new lines, so there is no sideways scrolling at 320 px. Status is shown in words and pattern. Time and length buttons are specified at 44 by 44 px, above the 24 by 24 px minimum, to be measured on the D4 prototype. The summary and the pass are written so a screen reader can announce them as status messages. On a phone today, the booking page is the desktop layout shrunk down, and you have to pan sideways to reach the calendar and the Book button (Figure A6).	3 to 5	Partly addressed. None of the six checkpoints has been checked yet; no assistive-technology user in our evidence (OR4).

4.3 Alternatives seriously considered and not selected

Alternative	Why we didn't select it
SA-1 Day timeline grid (runner-up)	It meets R2 as well as MY-1 on a wide screen and was the best at showing a whole day at once (P04-B2). We dropped it because a rooms-by-hours grid needs two-way scrolling at 320 px, which fails R5 checkpoint 1 (WCAG 2.2 reflow), and the D2 scenario happens on a phone (P01-C1, P08-C1). Its drag-toselect also brings back the click-and-undo problem students reported with the slider (S09-F2, S09-F3), which works against R4.
MU-2 Next free room	It is the fastest route for practiced bookers and is Full on R4. But it shows one room at a time, so a student can't identify every free and every taken room, which is exactly what R2's criterion tests. It only partly covers our leading requirement. We kept its best idea, showing the earliest free time first (P07-C2), by ordering each room's free times.
UH-1 Check-in and autorelease	It is the only alternative aimed at why the portal shows rooms as unavailable when they are empty (P09B1, SV-5) and at the double booking in S05-F1. We didn't select it because it is a library policy and backend change that we can't prototype or test in D4 and D5. We are passing it on as a recommendation to the library under open risk OR1.
MY-2 Floor-plan map	It matches how students check rooms by looking (P06-B1). It still needs a separate time step (R4), its room targets shrink at 320 px (R5 checkpoint 3), and we don't have a verified room layout to draw.

4.4 Why we selected it

MY-1 became the core because it is the only one of the 20 alternatives that could meet two D2 success criteria on its own, R2 and R4, and R2 carries the leading opportunity and POV1. It answers "which rooms are free?" on screen at journey stage 3, the step where students most visibly give up on the portal and ask people (P09-B2, P04-E1). And it does this in a single-column form that can pass R5 at phone width. SA-1 meets R2 just as well but fails R5 checkpoint 1 at 320 px; MU-2 meets R4 but can't pass R2's criterion.

We added MU-1 and MY-4 because nothing that changes the booking page touches R1 or R3, and each was the only Full rating in its column. MU-4's fixed length buttons went in because they remove the slider from the one choice MY-1 still left open. None of these changes MY-1's flow, so the result is still one direction rather than a bundle of features, and it stays inside the portal instead of adding a desk, kiosk or messaging service.

The need to know which rooms are free shows up at all three points on the persona spectrum, so the core concept serves personas A, B and C, not just one group. **4.5 Open risks**

ID	Risk	Req.	What D4 and D5 need to watch for
OR1	A clearer display can't fix wrong data. If the portal itself marks free rooms as unavailable (P09-B1, SV-5) or allows double booking (S05-F1), our design will just show the wrong status more clearly.	R2	R2's test uses prepared views, so it won't catch this. In D5, ask testers whether the status shown matched the real room. Recommend UH-1 (checkin and auto-release) to the library.

OR2	We can see room names and the numbers in brackets beside them, but we don't know the library's real rules: maximum length (P06-C3, "is there a limit?"), group size (P10-B1), how far ahead students can book, or what each calendar colour means.	R4, R3	Confirm with library staff before D4A. Until then, the 1, 2 and 3 hour buttons and the Today and Tomorrow tabs are our assumptions.
OR3	We never saw what the portal does after a booking is submitted. The pass assumes the portal gives back a reliable booking ID, and we haven't confirmed that it does.	R3	Prototype the pass and a clear success and failure state, and test R3 straight after a completed test booking, Check whether Insignia returns a booking reference before showing one as real.
OR4	R5 rests on WCAG checkpoints, two phone findings from single participants (P08, S06) and our own phone screenshots (Figure A6). No assistive-technology user was in our evidence. A long room list may also mean a lot of vertical scrolling.	R5	Run the six-checkpoint walkthrough on the D4 prototype at 320 px and 200% text, with at least one screen reader pass.
OR5	The home-page tile only helps students who open the portal. Persona A students who don't know booking exists (P01-B2) may never open it.	R1	R1's test starts from the portal address, so it won't show this. Awareness outside the portal (like SA-2's orientation email) is not part of the concept.
OR6	We haven't storyboarded the state where no room is free. Students currently respond by refreshing or asking people (P04-B1, P09-B2). The current portal also shows an empty room list with no explanation under some filter settings (Figure A4).	R2	D4A has to design this state. MU-5 (room watch) is the candidate to revisit.
OR7	Insignia is a third-party system, so we don't know whether its pages can be changed this way.	All	We test the concept as a prototype; implementation is out of scope.

Handoff to D4A. Prototype the flow as selected: find booking → see free rooms → pick time and length → review → booking succeeds or fails. Include the no-rooms-free state, a wrong-status state, and keyboard and screen-reader use. Any booking data in the prototype should be clearly marked as made up, and we should confirm the library's rules before treating lengths, booking IDs or limits as real.

Appendix A | Screenshots of the current portal

We took these screenshots on 22 October 2026, after the D1B sessions, to check that the problems we designed against are still present in the live portal. Figures A1 to A4 were taken in a desktop browser while signed in; Figures A5 and A6 were taken in a phone browser on the same website. They show what the portal displayed at that moment. They are not participant data, and they don't tell us whether any room was actually free.

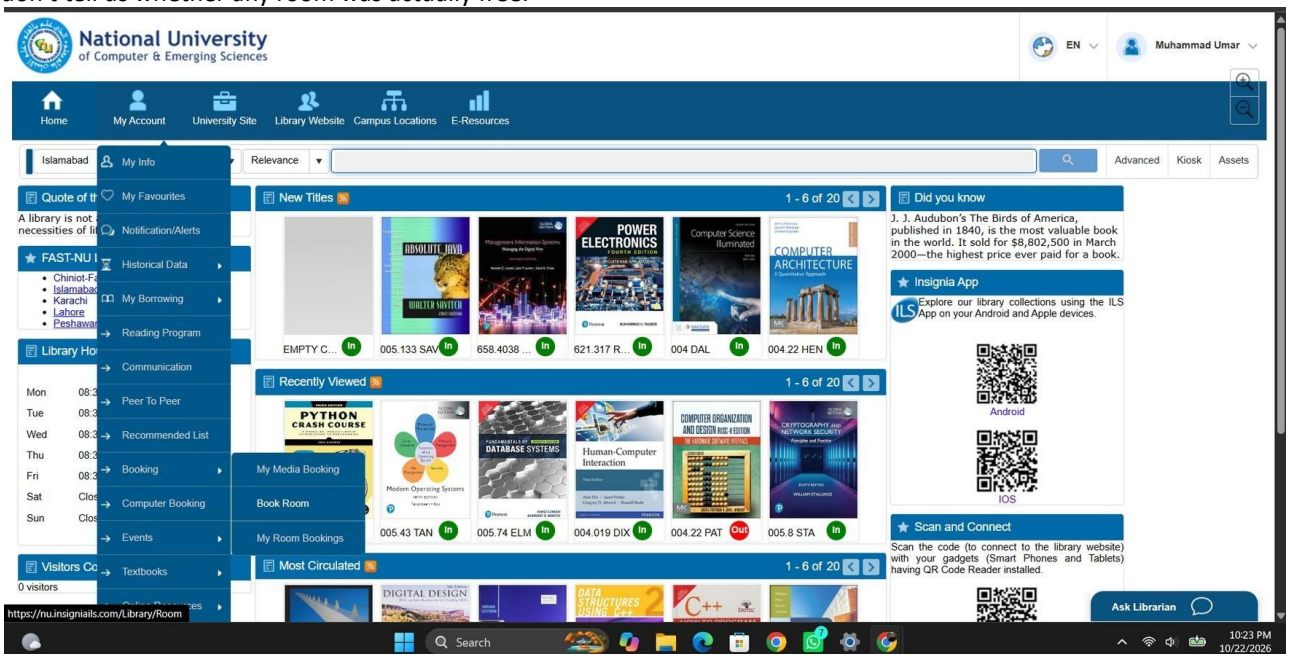


Figure A1. Desktop, signed in. Room booking is reached through My Account → Booking, which opens Book Room and My Room Bookings. There is no booking entry on the home page itself.

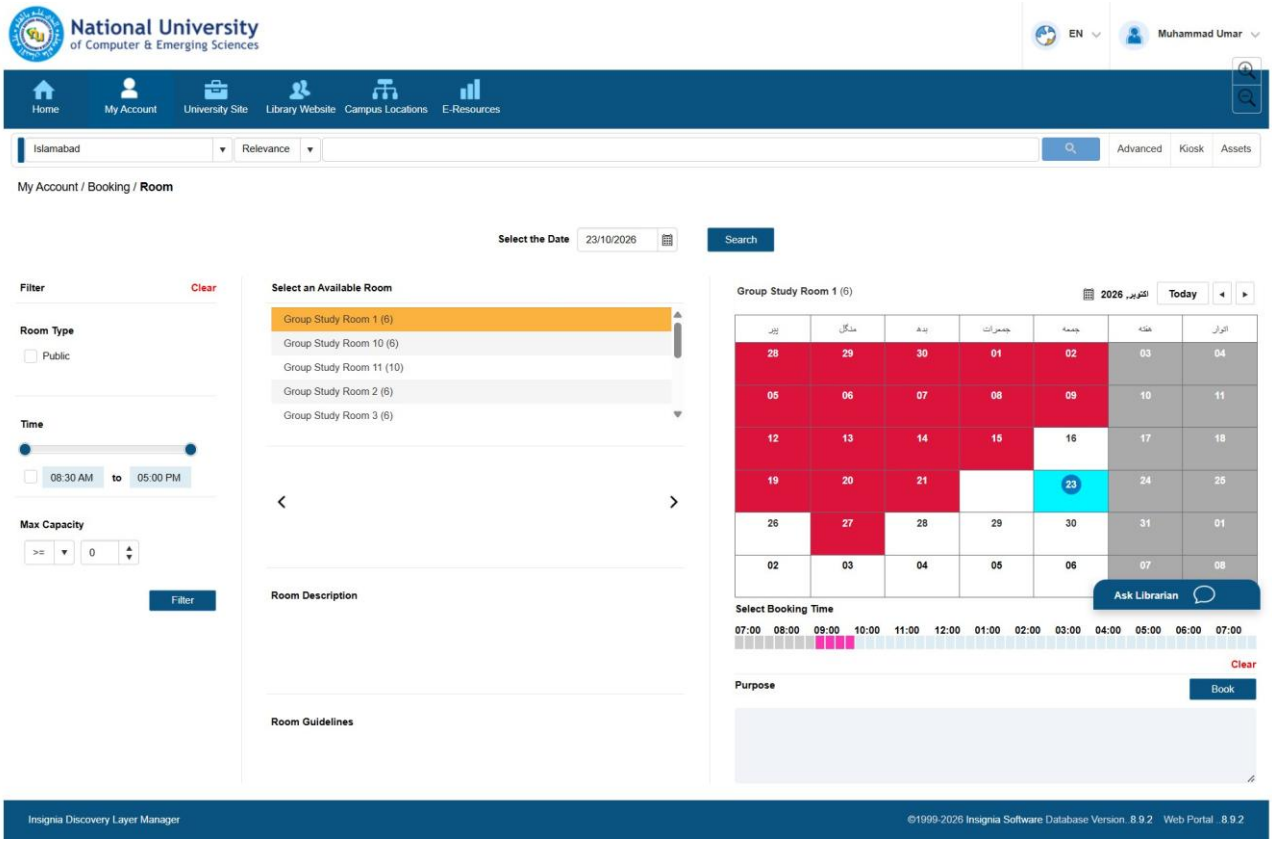


Figure A2. Desktop Book Room page for 23/10/2026: room list with a number in brackets after each name, a date field, a colour-coded month grid, a time-range slider under Filter, a separate Select Booking Time strip, a Purpose box and the Book button. We couldn't find a text legend for the calendar colours on this screen.

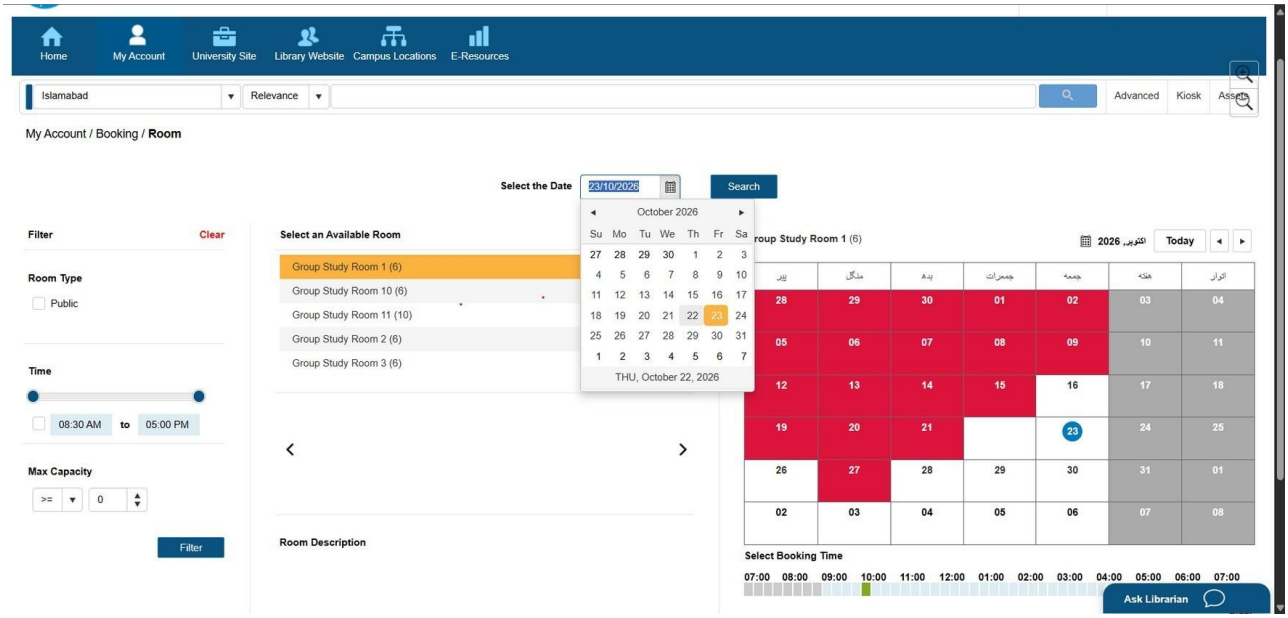


Figure A3. Desktop, same page with the date-picker popup open. There are two calendars on one screen: the popup for the date field, and the larger month grid beside it.

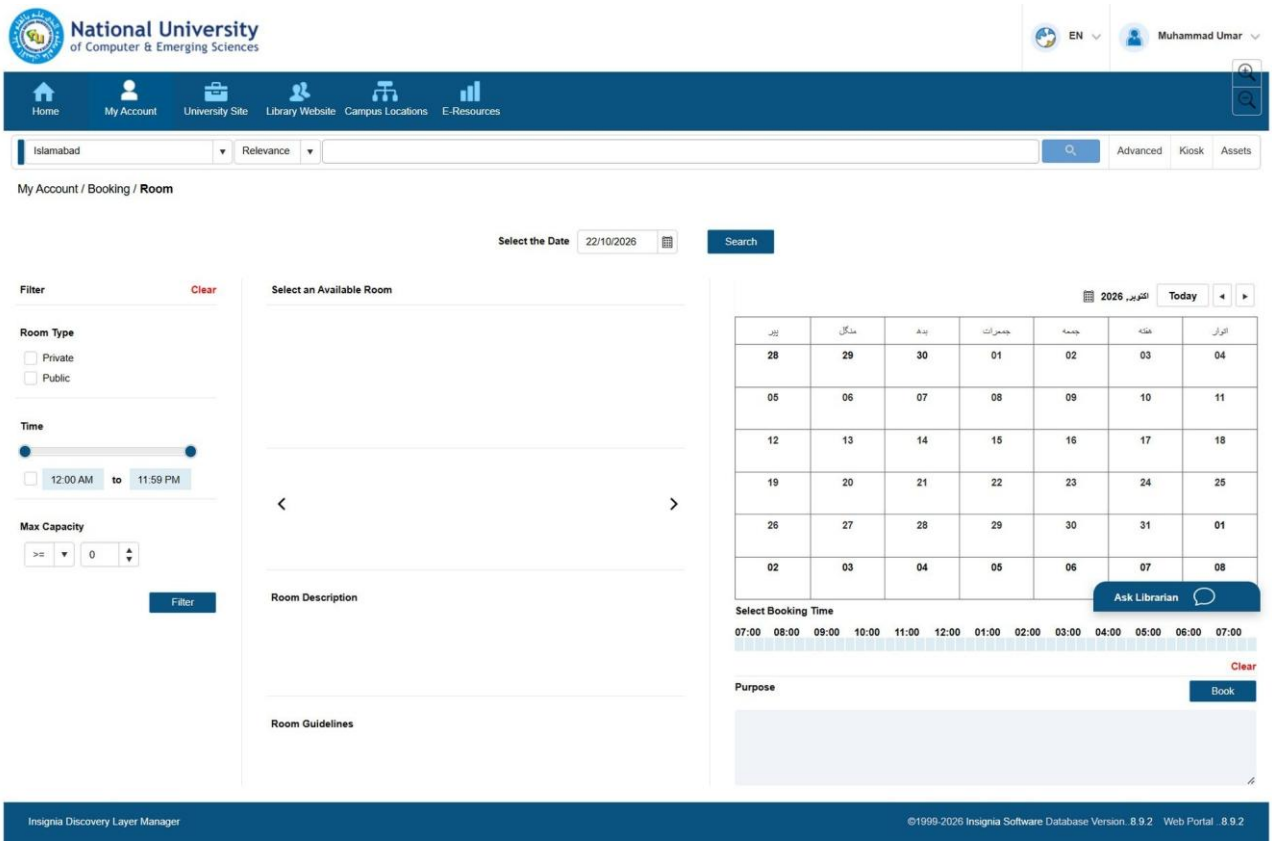


Figure A4. Desktop, 22/10/2026, with different filter settings from Figures A2 and A6: the Room Type list shows both Private and Public, and the time range is the whole day (12:00 AM to 11:59 PM). In this state the room list is empty and the month grid is uncoloured, with no message explaining why. Figure A6 shows rooms listed for the same date under the other filter settings, so we read this as the page giving no feedback when the filters return nothing, not as no rooms existing that day.

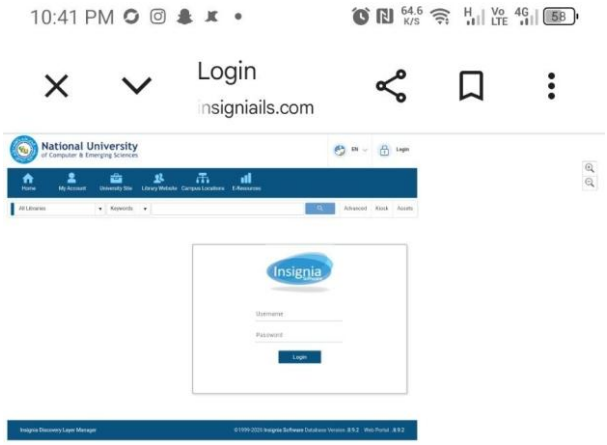


Figure A5. Phone browser, login page (cropped from the top of the screenshot). The desktop page is drawn small in the upper left, with tiny fields and a lot of empty space.

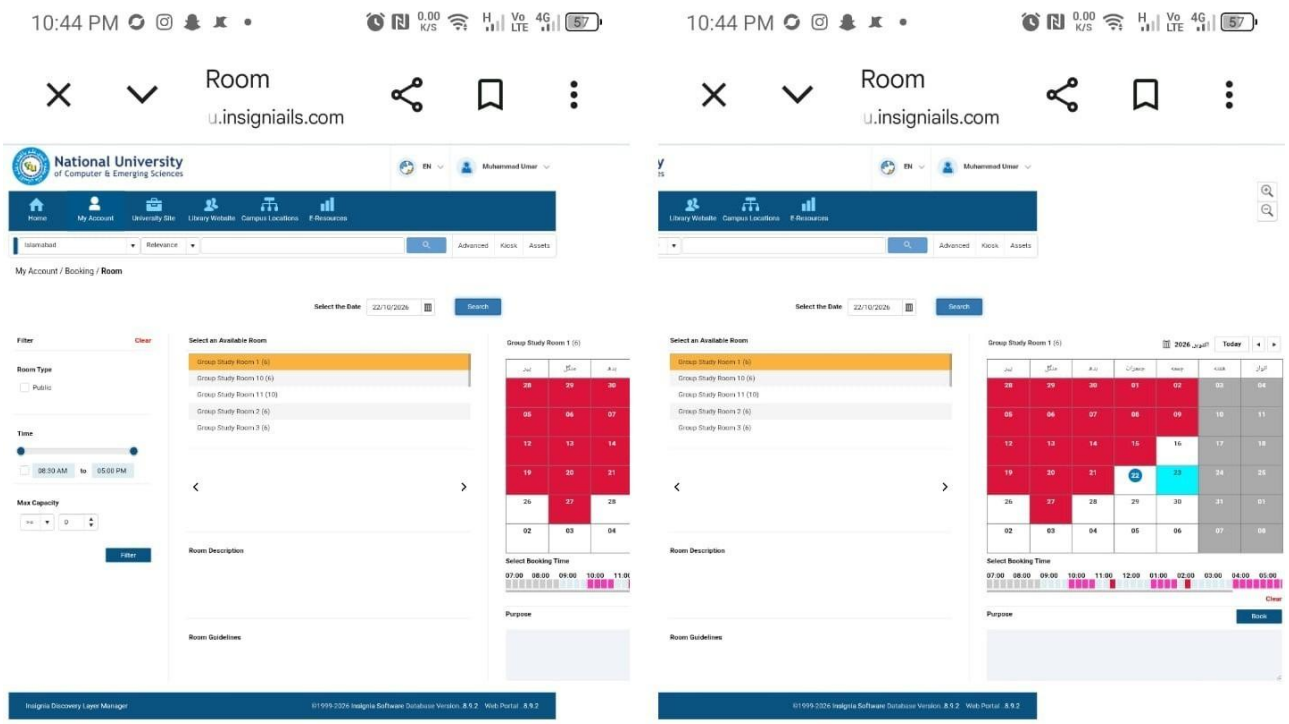


Figure A6. Phone browser, the same Book Room page in two screenshots side by side (blank lower parts cropped). Left: room list and filters. Right: after panning sideways, the calendar, time strip, Purpose box and Book button. The layout doesn't reflow to one column; it shrinks, and the booking controls are off-screen until you pan. This shows the R5 problem, but it doesn't measure contrast, target sizes, 200% text or screen-reader behaviour.